北师香港浸会大学
BEIJING NORMAL-HONG KONG BAPTIST UNIVERSITY

博文雅志 真知笃行
In knowledge and in deeds, unto the whole person

FIN3083: Financial Big Data Analytics

Group Project — Design a Novel Trading Strategy

Course:	FIN3083 – Financial Big Data Analytics
Due Date:	May 25, 2026 – 5:00 PM
Group Size:	4 students per group
Submission Format:	SAS code + written report
Presentation:	Week 14 (May 26, 2026)

1. Objective

The group project is the capstone assessment of FIN3083. Your task is to design, implement, and backtest a **novel trading strategy that is fundamentally different from those covered in this course** (MA, MACD, momentum, short-term reversal, and triangular arbitrage strategies). You will demonstrate mastery of the backtesting workflow, statistical rigor, and critical thinking about strategy design. All group members will present their work during Week 14.

2. Project Overview

Component	Description	Weight
SAS Implementation	Fully functional, well-commented SAS code implementing your strategy	40%
Written Report	8–12 page analysis of strategy design, methodology, results, and limitations	30%
Group Presentation	15-minute presentation with all members presenting (Week 14)	30%

3. Strategy Requirements

3.1 Novel Strategy Design

Your strategy must satisfy **all** of the following criteria:

3.1.1 Novelty

The strategy must be fundamentally different from MA, MACD, momentum, short-term reversal, and triangular arbitrage. Examples of acceptable novel approaches include:

- **Sentiment-Based Strategies:** Incorporate alternative data such as social media sentiment, news sentiment, or analyst recommendation changes to predict returns.
- **Machine Learning Approaches:** Use supervised learning (e.g., logistic regression, random forests, gradient boosting) to predict daily returns based on technical or fundamental features.
- **Pairs Trading / Statistical Arbitrage:** Identify pairs of stocks with historically correlated returns and trade mean-reversions in their spread.
- **Volatility-Based Strategies:** Design a strategy that exploits volatility clustering or volatility mean-reversion.
- **Microstructure-Based Strategies:** Exploit order flow imbalances, bid-ask spreads, or volume patterns.
- **Multi-Factor Strategies:** Combine multiple factors (value, momentum, quality, low volatility).
- **Event-Driven Strategies:** Respond to earnings announcements, dividends, M&A, etc.
- **Regime-Switching Strategies:** Adapt signals based on market regimes.

3.1.2 Theoretical Justification

Provide a clear narrative explaining **why** your strategy should work. Reference academic papers or theory whenever necessary.

3.1.3 Data Availability

Use BNBU subscribed or publicly available data (CSMAR, CRSP, Compustat, Kaiko, Yahoo Finance, Coinmarketcap).

3.1.4 Implementability

Must be backtestable on at least: - 5 years (high-frequency strategy) - 20 years (low-frequency strategy)

3.2 Backtesting Methodology

Your implementation must follow industry-standard backtesting practices:

- **Look-Ahead Bias Prevention:** Ensure that no future information leaks into past trading decisions.
- **Transaction Costs:** Incorporate realistic trading costs (e.g., bid-ask spreads, commissions, market impact) into your returns calculation.
- **Survivorship Bias:** If using stock universes, account for delisted or newly listed stocks.
- **Out-of-Sample Validation:** Split your data into training (in-sample) and testing (out-of-sample) periods to avoid overfitting.

3.3 Performance Metrics

Metric	Definition	Interpretation
Total Return	Cumulative return over the backtest period	Absolute profitability
Annualized Return	Geometric mean return scaled to annual frequency	Risk-adjusted profitability
Sharpe Ratio	(Return - Risk-Free Rate) / Volatility	Risk-adjusted return per unit of risk
Sortino Ratio	(Return - Risk-Free Rate) / Downside Volatility	Risk-adjusted return focusing on downside risk
Maximum Drawdown	Largest peak-to-trough decline	Worst-case loss scenario
Calmar Ratio	Annualized Return / Maximum Drawdown	Return per unit of drawdown risk
Win Rate	Percentage of profitable trading days or trades	Consistency of strategy

Additionally, provide **at least two custom metrics** relevant to your specific strategy (e.g., average holding period, trade frequency, correlation with market, beta, etc.).

4. Deliverables

4.1 SAS Code

Submit a well-organized SAS program including:

- Data preparation
- Feature engineering
- Signal generation
- Backtesting
- Metrics + visualization

4.1.1 Code Quality Standards

- Clear comments
- Descriptive variable names
- Efficient structure
- Must run end-to-end without errors

4.2 Written Report

Your report should be 8–12 pages (single-spaced, 11pt font) and include the following sections:

1. **Executive Summary (1 page):** Concise overview of your strategy, key results, and main insights.
2. **Introduction & Motivation (1–2 pages):** Explain the financial problem your strategy addresses. What market inefficiency or anomaly does it exploit? Cite relevant academic research or market observations.
3. **Strategy Design & Methodology (2–3 pages):** Describe your strategy in detail:
    - What are the trading signals?
    - How are entry and exit points determined?
    - What is the holding period?
    - How do you manage risk (stop-losses, position sizing, etc.)?
    - How does your approach differ from existing strategies?

4. **Data & Implementation (1–2 pages):** Document your data sources, sample period, universe of stocks, and any data preprocessing steps. Explain how you handled missing data, corporate actions, or survivorship bias.

5. **Backtesting Results (2–3 pages):** Present your performance metrics, equity curves, and drawdown analysis. Compare your strategy to a buy-and-hold benchmark. Discuss in-sample vs. out-of-sample performance.

6. **Robustness & Sensitivity Analysis (1–2 pages):** Test your strategy across different market regimes, time periods, or parameter configurations. How robust is your strategy to changes in assumptions?

7. **Limitations & Future Work (1 page):** Critically discuss the limitations of your backtest (e.g., slippage, liquidity constraints, parameter overfitting). What would be needed to deploy this strategy in live trading?

8. **References:** Cite all academic papers, data sources, and other materials referenced in your report.

4.3 Group Presentation

Each group will deliver a **15-minute presentation** during Week 14 (May 26, 2026). **All group members must participate in the presentation.** The presentation should cover:

- **Strategy Overview (2–3 minutes):** What is your strategy? Why is it novel and theoretically justified?
- **Methodology (3–4 minutes):** How does it work? What are the key signals and decision rules?
- **Results (4–5 minutes):** What are your key performance metrics? How does it compare to buy-and-hold? Include equity curve and drawdown visualizations.
- **Robustness & Limitations (2–3 minutes):** How robust is your strategy? What are its key limitations?
- **Q&A (1–2 minutes):** Be prepared to answer questions from the instructor and peers.

4.3.1 Presentation Guidelines

- Use clear, professional slides (Latex, PowerPoint, Google Slides, or similar)
- Include visualizations (equity curves, drawdown charts, signal distributions)
- Ensure all members contribute meaningfully to the presentation
- Practice your timing to stay within 15 minutes

5. Submission Guidelines

Submit the following files via the course portal:

1. **SAS Code:** `GroupProject_<GroupNumber>.sas` (or multiple files: `GroupProject_<GroupNumber>_` etc.)
2. **Written Report:** `GroupProject_<GroupNumber>_Report.pdf`

Example: Group 5 becomes `GroupProject_5.sas`, `GroupProject_5_Report.pdf`

5.1 Required Header Comment Block (SAS Code)

```
/************************************************************
FIN3083 Group Project: Novel Trading Strategy Design
Group Number: [Your Group Number]
Group Members:
Name1 (ID1): please specify responsibility here
Name2 (ID2): please specify responsibility here
Name3 (ID3): please specify responsibility here

Strategy Name: [Your Strategy Name]
Strategy Description: [2-3 sentence summary]

--- Key Results ---
Total Return: [%]
Annualized Return: [%]
Sharpe Ratio: [Value]
Sortino Ratio: [Value]
Maximum Drawdown: [%]
Calmar Ratio: [Value]
Win Rate: [%]
Profit Factor: [Value]

--- Custom Metrics ---
Metric 1: [Name] = [Value]
Metric 2: [Name] = [Value]

--- Notes ---
[Any important observations or caveats]
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```

6. Grading Criteria

6.1 SAS Implementation (40%)

Criterion	**Weight**	**Description**
Code Correctness	50%	Does the code run without errors? Are all required steps implemented correctly?
Backtesting Rigor	25%	Are transaction costs, look-ahead bias, and survivorship bias properly handled?
Performance Metrics	15%	Are all required metrics calculated correctly?
Code Quality	10%	Is the code well-organized, readable, and well-commented?

6.2 Written Report (30%)

Criterion	**Weight**	**Description**
Strategy Novelty	20%	Is the strategy novel? Is it well-justified theoretically?
Clarity	15%	Is the report well-structured and easy to follow?
Methodology	25%	Is the backtesting methodology sound? Are limitations acknowledged?
Results	25%	Are results clearly presented? Do you provide meaningful interpretation?
Writing Quality	15%	Is the writing professional and free of errors?

6.3 Group Presentation (30%)

Criterion	**Weight**	**Description**
Content	40%	Is the strategy clearly explained? Are key results presented effectively?
Engagement	30%	Do all members participate? Is the presentation engaging and well-paced?
Visual Quality	20%	Are slides professional and visually clear? Do visualizations support the narrative?
Questions	10%	Can the group answer questions confidently and accurately?

7. Strategy Uniqueness Verification

To ensure that strategies across different groups are sufficiently different, each group must submit their **project title and a 2–3 sentence strategy description** by **April 27, 2026**. The instructor will use an AI-based similarity checker to verify that no two groups are pursuing substantially similar strategies. Groups with highly similar strategies will be required to modify their approach.

Submission Portal: Groups will submit their titles and descriptions via a dedicated web portal that provides real-time feedback on strategy uniqueness.

8. Key Dates and Milestones

Milestone	**Date**	**Deliverable**
Group Formation	April 27, 2026	Group roster + strategy
Mid-Project Checkpoint	May 11, 2026	Draft code
Final Submission	May 25, 2026	Complete project
Group Presentations	May 26, 2026	Presentation

9. Support and Resources

- **Course Materials:** All lecture slides and code.

- **SAS Documentation:** PROC EXPAND, PROC SQL, PROC SGPLOT, and other relevant procedures.
- **Databases FBM Subscribed:** https://fbm.bnbu.edu.cn/research/database.htm

10. Academic Integrity

All work submitted must be original and produced by your group. You are allowed to use generative AI tools. However, any external code, data, or ideas must be properly reported in your report. Plagiarism or academic dishonesty will result in a zero grade for the entire group project and potential disciplinary action.

11. Final Notes

This group project is an opportunity to demonstrate your mastery of financial big data analytics, backtesting methodology, and critical thinking. Be creative, rigorous, and ambitious in your strategy design. The best projects will combine novelty, theoretical soundness, empirical validation, and clear communication.

Good luck!

Due: May 25, 2026 — 5:00 PM